

# Jiaming Lu

jiaming.lu.0124@gmail.com | [github.com/jiamingluuu](https://github.com/jiamingluuu) | (+86) 173-2686-6550 | [linkedin.com/in/jiaming-lu-924992253/](https://www.linkedin.com/in/jiaming-lu-924992253/)

Student Pass Holder | Available Immediately

## Education

Nanyang Technological University

2026.08 – 2027.11 (Expected)

- MSc in Artificial Intelligence

University of Toronto

2021.09 – 2026.06

- BSc in Computer Science

## Work Experience

TikTok | Software Engineer Intern

2026.05 – 2026.12 (Expected)

- Drove a production search-ranking architecture migration, moving 120+ query- and user-side features across the coarse- and fine-ranking stages.
- Designed, implemented, and validated an automated differential-testing pipeline spanning five microservices, comparing outputs from the legacy and new architectures to ensure feature parity throughout the migration.
- Reworked the fine-ranking operator pipeline for dual-path query-correction traffic, reducing the average predicted CTR score gap between the legacy and new services by 8% and ultimately achieving metric parity.

Tencent | Software Engineer Intern

2025.05 – 2025.08

- Optimized an alert root-cause analysis agent, saving 15+ engineering hours per week with 87% diagnostic accuracy.
- Designed a human-in-the-loop feedback pipeline to capture failure cases and refine prompts/workflows, enhancing alert resolution accuracy and accelerating iteration.
- Built an asynchronous proxy layer with thread-safe LRU-K cache for a user profile computation system, cutting storage cluster network bandwidth consumption by 43% and reducing read amplification by 79% for upstream RPCs.
- Implemented a cache-aware SIMD pooling operator for embedding pooling, reducing TP99 latency from 264μs to 93μs.

Urban Data Centre | AI/NLP Analyst

2024.09 – 2025.03

- Built a RAG pipeline for entity disambiguation in homonymous references, reducing hallucinated answer generations.
- Developed Python ETL pipelines to transform complex knowledge graph ontologies into XML/JSONL corpora, enabling a robust knowledge source for the RAG-based question answering.
- Fine-tuned a LLaMA-3-8B model with LoRA for English-to-SPARQL generation, replacing a rule-based baseline and improving query precision using schema-aware prompting.

JD.com, Inc. | Software Engineer Intern

2024.05 – 2024.08

- Contributed to backend middleware for JD's shopping navigation search services on a large-scale e-commerce platform.
- Engineered a thread-safe ETL pipeline with rate limiting for incremental index loading, enabling real-time search updates.
- Refactored service routing registration using Singleton Pattern (500+ LoC), eliminating duplicated configuration.
- Authored technical documentation for modern C++ practices, adopted as a cross-team reference for intern onboarding.

University of Toronto | Teaching Assistant

2023.01 – 2026.04

- Delivered grading, office hours, and one-on-one tutoring for undergraduate computer science and mathematics courses.
- Supported 300+ students and provided 420+ hours of tutoring across Introduction to Computer Science and Linear Algebra.

## Publications

- Lu, J. (2024). Parallel implementation of Wiener's attack on RSA: Algorithm design and performance evaluation. *Applied and Computational Engineering*, 37 184–190. <https://doi.org/10.54254/2755-2721/37/20230503>

## Technical Skills

- **Programming Languages:** C, C++, Go, Python, Rust, Java, JavaScript, Bash, SQL.
- **Frameworks/Libraries:** PyTorch, TensorFlow, Hugging Face Transformers, PEFT, scikit-learn, LangChain, LangGraph.
- **Tools/Systems:** Linux, Git, Docker, Redis, Kafka, MySQL, PostgreSQL, AddressSanitizer, Valgrind, pprof.
- **Languages:** English (Advanced), Chinese (Native).